

PROGRAM 1:

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, a 'Secure Exit' button, the lab name 'SIMATS Online Programming Lab', the subject 'JAVA PROGRAMMING', and a user profile for MULE MADHU SNEHA LATHA REDDY. The left sidebar has a dropdown menu set to 'Arithmetic' and a description: 'Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.' The main editor area contains the following Java code:

```
1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         int a;
5         int b;
6         Scanner sc=new Scanner(System.in);
7         System.out.println("enter two numbers");
8         a=sc.nextInt();
9         b=sc.nextInt();
10        System.out.println("add="+(a+b));
11        System.out.println("sub="+(a-b));
12        System.out.println("mul="+a*b);
13        System.out.println("div="+a/b);
14    }
15 }
```

Below the code editor is an 'OUTPUT' section showing the program's execution results:

```
enter two numbers
add=11
sub=1
mul=30
div=1
```

On the right side, there is an input field with the numbers 6 and 5.

PROGRAM 2:

The screenshot shows the SIMATS Online Programming Lab interface. The header is identical to the first screenshot. The left sidebar has a dropdown menu set to 'MAX' and a description: 'Write a Java program to find the maximum of three numbers using conditional statements.' The main editor area contains the following Java code:

```
1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         int a, b, c;
5         Scanner sc=new Scanner(System.in);
6         System.out.println("enter three numbers");
7         a=sc.nextInt();
8         b=sc.nextInt();
9         c=sc.nextInt();
10        if(a>b)
11            System.out.println("a is maximum");
12        else if(b>c)
13            System.out.println("b is maximum");
14        else
15            System.out.println("c is maximum");
16    }
17 }
```

Below the code editor is an 'OUTPUT' section showing the program's execution results:

```
enter three numbers
b is maximum
```

On the right side, there is an 'INPUT' section with the numbers 30, 34, and 29.

PROGRAM 3:



SIMATS Online Programming Lab

MULE MADHU SNEHA LATHA REDDY
192425092RD

QUESTIONS 28 / 31 completed

Leap

LEAP
Write a Java program to check whether a given year is a leap year or not using conditional statements.

PROGRAM EDITOR

Java
Run Save

```

1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         int a;
5         Scanner sc=new Scanner(System.in);
6         System.out.println("enter a year");
7         a=sc.nextInt();
8         if (a%4==0)
9             System.out.println("it is leap year");
10        else
11            System.out.println("it is not a leap year");
12        }
13    }

```

OUTPUT

```

enter a year
it is not a leap year

```

INPUT

```

2025

```

PROGRAM 4:



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QUESTIONS 28 / 31 completed

Fibonacci

FIBONACCI
Write a Java program to print the Fibonacci series up to a given number using loops.

PROGRAM EDITOR

Java
Run Save

```

1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args) {
4
5         Scanner sc=new Scanner(System.in);
6         int n, b=1,a=0,c;
7         System.out.println("enter a values");
8         n = sc.nextInt();
9         System.out.println("Fibonacci Series: ");
10
11        while (a <= n){
12            System.out.print(a+" ");
13            c=a+b;
14            a=b;
15            b=c;
16        }
17        sc.close();
18    }
19 }

```

OUTPUT

```

enter a values
Fibonacci Series:
0 1 1 2 3 5 8 13

```

INPUT

```

20

```

PROGRAM 5:


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192425092RD

JAVA PROGRAMMING

Factorial

FACTORIAL

Write a Java program to calculate the factorial of a given number using loops.

```

1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         int n;
5         Scanner sc=new Scanner(System.in);
6         System.out.println("enter a number");
7         n=sc.nextInt();
8         int fact=1;
9         for(int i=1; i<=n; i++){
10            fact=fact*i;
11        }
12        System.out.println("factorial="+fact);
13    }
14 }

```

5

OUTPUT

```

enter a number
factorial=120

```

PROGRAM 6:


SIMATS Online Programming Lab


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192425092RD

JAVA PROGRAMMING

RT-Pattern

RT-PATTERN

Write a Java program to print the following pattern using loops:
**


```

1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4
5         Scanner sc=new Scanner(System.in);
6         int n=sc.nextInt();
7         for(int i=1; i<=n; i++){
8             for(int j=1; j<=i; j++){
9                 System.out.print(" ");
10            }
11            System.out.println();
12        }
13    }
14 }

```

5

OUTPUT

```

*
* *
* * *
* * * *
* * * * *

```

PROGRAM 7:



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192425092RD

JAVA PROGRAMMING

Search

SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

```

1 import java.util.Scanner;
2 class Main{
3 public static void main(String[] args){
4     Scanner sc=new Scanner(System.in);
5     int n=sc.nextInt();
6     int[] a=new int[n];
7     for(int i=0; i<n; i++)
8         a[i]=sc.nextInt();
9     int key=sc.nextInt();
10
11     int low=0,high=n-1;
12
13     while(low <= high) {
14         int mid=(low+high)/2;
15         if(a[mid]==key){
16             System.out.println("found");
17             return;
18         } else if(a[mid]<key) {
19             low = mid+1;
20         } else{
21             high=mid-1;

```

OUTPUT

found

5

10 20 30 40 50

30

PROGRAM 8:



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192425092RD

JAVA PROGRAMMING

GCD

GCD

Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

```

1 import java.util.Scanner;
2 class Main{
3 public static void main(String[] args){
4     int a,b;
5     Scanner sc= new Scanner(System.in);
6     System.out.println("enter a number");
7     a=sc.nextInt();
8     b=sc.nextInt();
9     int gcd=1;
10    for(int i=1; i<=a && i<=b; i++){
11        if(a%i==0 && b%i==0){
12            gcd=i;
13        }
14    }
15    System.out.print("gcd="+gcd);
16 }
17 }

```

OUTPUT

enter a number
gcd=12

24

36

PROGRAM 9:

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192425092RD

Prime

PRIME
Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

```

1 import java.util.Scanner;
2 class main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         System.out.println("enter a number");
6         int n=sc.nextInt();
7         int count=0;
8         for(int i=1; i<=n; i++){
9             if(n%i==0){
10                 count++;
11             }
12         }
13         if(count==2){
14             System.out.println(n+ " it is a prime number");
15         }else{
16             System.out.println(n+ " it is not a prime number");
17         }
18     }
19 }

```

OUTPUT

```

enter a number
7
it is a prime number

```

7

PROGRAM 10:

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192425092RD

Sort

SORT
Write a Java program to implement selection sort on an array of integers using loops and conditional statements.

```

1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4
5         Scanner sc=new Scanner(System.in);
6
7         int[] arr=new int[5];
8         for(int i=0; i<5; i++){
9             arr[i] = sc.nextInt();
10        }
11        for(int i=0; i<4; i++){
12            int min=i;
13            for(int j=i+1; j<5;j++){
14                if(arr[j]<arr[min]){
15                    min=j;
16                }
17            }
18            int temp=arr[i];
19            arr[i]=arr[min];
20            arr[min]=temp;
21        }

```

OUTPUT

```

11 12 22 24 64

```

64 24 12 22 11

PROGRAM 11:



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192425092RD

Table

TABLE
Write a Java program to print the multiplication table of a given number using loops.

```

1 import java.util.Scanner;
2 class main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int n=sc.nextInt();
6         for(int i=1; i<=10; i++){
7             System.out.println(n + " x " + i + " = " + (n * i));
8         }
9     }
10 }
11

```

OUTPUT

```

3 x 1 = 3
3 x 2 = 6
3 x 3 = 9

```

3

PROGRAM 12:



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192425092RD

Factorial

FACTORIAL
2. Write a Java program to find the factorial of a given number using recursion.

```

1 import java.util.Scanner;
2 public class Factorial{
3     static int fact(int n){
4         if (n<=1)
5             return 1;
6         return n* fact(n-1);
7     }
8     public static void main(String[] args){
9         Scanner sc = new Scanner(System.in);
10        int n= sc.nextInt();
11        System.out.println(fact(n));
12    }
13 }

```

OUTPUT

```

720

```

6

PROGRAM 13:



SIMATS Online Programming Lab

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192425092RD

LT-Pattern

LT-PATTERN
Write a Java program to print the following pattern using loops:*****

*

```

1 import java.util.Scanner;
2
3 class Main{
4     public static void main(String[] args){
5         int n;
6         Scanner sc=new Scanner(System.in);
7         n=sc.nextInt();
8         for(int i=n; i>=1; i--){
9             for(int j=1; j<=i; j++){
10                 System.out.print("***");
11             }
12             System.out.println();
13         }
14     }
15 }

```

OUTPUT

```

*****
*****
***
*

```

5

PROGRAM 14:



SIMATS Online Programming Lab

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192425092RD

LCM

LCM
Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

```

1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5         int a=s.nextInt();
6         int b=s.nextInt();
7
8         int max=(a>b)?a:b;
9         while(true){
10             if (max%a==0 && max%b==0){
11                 System.out.println(max);
12                 break;
13             }
14             max++;
15         }
16     }
17 }

```

OUTPUT

```

36

```

12

18

PROGRAM 15:



SIMATS Online Programming Lab

JAVA PROGRAMMING

MULE MADHU SNEHA LATHA

REDDY

192425092RD

QUESTIONS 28 / 31 completed

D to B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```

1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5
6         int decimal=sc.nextInt();
7         int binary=0;
8         int base=1;
9
10        while(decimal>0){
11            int rem=decimal%2;
12            binary=binary+(rem*base);
13            base=base*10;
14            decimal=decimal/2;
15        }
16        System.out.println(binary);
17    }
18 }

```

OUTPUT

1010

INPUT

10

PROGRAM 16:



SIMATS Online Programming Lab

JAVA PROGRAMMING

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REDDY

192425092RD

QUESTIONS 28 / 31 completed

B to D

Write a Java program to convert a binary number to decimal using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```

1 import java.util.Scanner;
2
3 class Main{
4     public static void main(String[] args){
5         Scanner sc=new Scanner(System.in);
6
7         int binary=sc.nextInt();
8         int decimal=0;
9         int base=1;
10
11        while(binary>0){
12            int rem=binary%10;
13            if(rem==1){
14                decimal=decimal+base;
15            }
16            base=base*2;
17            binary=binary/10;
18        }
19        System.out.println(decimal);
20    }
21 }

```

OUTPUT

10

INPUT

1010

PROGRAM 17:



SIMATS Online Programming Lab
JAVA PROGRAMMING

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192425092RD

QUESTIONS 28 / 31 completed

Sum

SUM
Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1
2 import java.util.Scanner;
3
4 class Main{
5     public static void main(String[] args){
6         Scanner sc= new Scanner(System.in);
7
8         int num=sc.nextInt();
9         int sum=0;
10        while(num>0){
11            int rem=num%10;
12            sum=sum+rem;
13            num=num/10;
14        }
15        System.out.println(sum);
16    }
17 }
18
```

OUTPUT
10

INPUT
1234

PROGRAM 18:



SIMATS Online Programming Lab
JAVA PROGRAMMING

MULE MADHU SNEHA LATHA REDDY
192425092RD

QUESTIONS 28 / 31 completed

Product

PRODUCT
Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5
6         int dig=sc.nextInt();
7         int prod=1;
8
9         while(dig>0){
10             int rem=dig%10;
11             prod=prod*rem;
12             dig=dig/10;
13         }
14         System.out.println(prod);
15     }
16 }
```

OUTPUT
24

INPUT
1234

PROGRAM 19:



SIMATS Online Programming Lab

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192425092RD

QUESTIONS 28 / 31 completed

Power

POWER
Write a Java program to find the power of a number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```

1
2 import java.util.Scanner;
3 class Main{
4     public static void main(String[] args){
5         Scanner sc=new Scanner(System.in);
6
7         int num=sc.nextInt();
8         int power=sc.nextInt();
9         int result=1;
10
11        while(power>0){
12            result=result*num;
13            power--;
14        }
15        System.out.println(result);
16    }
17 }

```

OUTPUT

8

INPUT

2
3

PROGRAM 20:



SIMATS Online Programming Lab

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192425092RD

QUESTIONS 28 / 31 completed

Area Circle

AREA CIRCLE
Write a Java program to calculate the area of a circle using the radius input by the user.

PROGRAM EDITOR

Java

Run

Save

```

1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         Scanner sc= new Scanner(System.in);
5         int r=sc.nextInt();
6         System.out.println("enter the radius");
7         double pi=3.14;
8         System.out.println(pi*r*r);
9     }
10 }

```

OUTPUT

enter the radius
78.5

INPUT

5

PROGRAM 21:

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 JAVA PROGRAMMING

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 192425092RD

Shpere

SHPERE

 Write a Java program to calculate the volume of a sphere using the radius input by the user.

```

1 import java.util.Scanner;
2 class Main{
3   public static void main(String[] args){
4     Scanner sc=new Scanner(System.in);
5     int r=sc.nextInt();
6     System.out.println("enter the radius");
7     double pi=3.14;
8     double volume = ((4.0/3.0)*pi*r*r*r);
9     System.out.println("volume of a sphere: "+volume);
10  }
11 }

```

 OUTPUT


```

enter the radius
volume of a sphere: 523.3333333333334

```

5

PROGRAM 22:

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SIMATS Online Programming Lab
 JAVA PROGRAMMING

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 192425092RD

QUESTIONS 28 / 31 completed

Palindrome

PALINDROME

 Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

PROGRAM EDITOR
 Java
 Run Save

```

1 import java.util.Scanner;
2 class Main{
3   public static void main(String[] args){
4     Scanner s=new Scanner(System.in);
5     String str=s.nextline();
6     String rev = "";
7
8     for(int i=str.length()-1; i>=0; i--){
9       rev = rev + str.charAt(i);
10    }
11    if(str.equals(rev)) {
12      System.out.println("palindrome");
13    } else {
14      System.out.println("not palindrome");
15    }
16  }
17 }
18 }

```

OUTPUT


```

palindrome

```

 INPUT

 madam

PROGRAM 23:

SIMATS Online Programming Lab

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192425092RD

QUESTIONS: 28 / 31 completed

Armstrong

ARMSTRONG
Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

PROGRAM EDITOR

Java
Run
Save

```

1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int num=sc.nextInt();
6         int a=num;
7         int sum=0;
8
9
10        while(a>0){
11            int rem=a%10;
12            sum=sum+(rem*rem*rem);
13            a=a/10;
14        }
15        if(sum==num){
16            System.out.println("Armstrong number");
17        }else{
18            System.out.println("not armstrong number");
19        }
20    }
21 }

```

OUTPUT
Armstrong number

INPUT
153

PROGRAM 24:

SIMATS Online Programming Lab

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192425092RD

RT-num.Pattern

**RT-
NUM.PATTERN**
Write a Java program to print the following pattern using loops:
1
22
333
4444
55555

PROGRAM EDITOR

Java
Run
Save

```

1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4
5         Scanner sc=new Scanner(System.in);
6         int n=sc.nextInt();
7         for(int i=1; i<=n; i++){
8             for(int j=1; j<=i; j++){
9                 System.out.print(i);
10            }
11            System.out.println(" ");
12        }
13    }
14 }

```

OUTPUT
1
22
333
4444

INPUT
5

PROGRAM 25:

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ENGINEERING

SIMATS Online Programming Lab
 JAVA PROGRAMMING

MULE MADHU SNEHA LATHA REDDY
 192425092RD

Perfect

PERFECT
 Write a Java program to check whether a given number is a perfect number or not using loops and conditional statements.

```

1 import java.util.Scanner;
2
3 class Perfectnumber{
4   public static void main(String[] args){
5     Scanner sc=new Scanner(System.in);
6
7     int num=sc.nextInt();
8     int sum=0;
9     for(int i=1; i<num; i++)
10      if(num%i==0){
11        sum+=i;
12      }
13     if(sum==num){
14       System.out.println("it is perfect number");
15     }
16     else{
17       System.out.println("it is not a perfect number");
18     }
19   }
20 }

```

28

OUTPUT
 it is prime number

PROGRAM 26:

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 JAVA PROGRAMMING

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 192425092RD

Max Array

MAX ARRAY
 Write a Java program to find the largest element in an array of integers using loops and conditional statements.

```

1 import java.util.Scanner;
2 class Main{
3   public static void main(String[] args){
4     Scanner sc=new Scanner(System.in);
5
6     int n=sc.nextInt();
7     int a[]=new int[n];
8
9     for(int i=0; i<n; i++)
10      a[i]=sc.nextInt();
11     int max=a[0];
12     for(int i=1; i<n; i++){
13       if(max<a[i])
14         max=a[i];
15     }
16     System.out.println(max);
17   }
18 }

```

5
 10 23 76 12 36

OUTPUT
 76

PROGRAM 27:



SIMATS Online Programming Lab
JAVA PROGRAMMING

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192425092RD

2nd Max

2ND MAX
Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         Scanner s=new Scanner(System.in);
5
6         int n=s.nextInt();
7         int a[]=new int[n];
8
9         for(int i=0;i<n; i++){
10             a[i]=s.nextInt();
11             int max=a[0], second=a[1];
12             for(int i=1; i<n; i++){
13                 if(max<a[i]){
14                     second=max;
15                     max=a[i];
16                 }
17                 else if(a[i]>second && a[i]!=max){
18                     second=a[i];
19                 }
20             }
21             System.out.println(second);
22         }
```

OUTPUT

78

5

12 67 89 36 78

PROGRAM 28:



SIMATS Online Programming Lab
JAVA PROGRAMMING

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QUESTIONS 29 / 31 completed

Min

MIN
Write a Java program to find the smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```
1 import java.util.Scanner;
2
3 class Main{
4     public static void main(String[] args){
5         Scanner sc=new Scanner(System.in);
6
7         int n=sc.nextInt();
8         int a[]=new int[n];
9
10        for(int i=0; i<n; i++){
11            a[i]=sc.nextInt();
12            int min=a[0];
13            for(int i=1; i<n; i++){
14                if(min>a[i])
15                    min=a[i];
16            }
17            System.out.println(min);
18        }
19    }
```

OUTPUT

18

5

93 26 18 39 22

PROGRAM 29:

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and a user profile for MULE MADHU SNEHA LATHA REDDY. The left sidebar shows 'QUESTIONS 31 / 31 completed' and a '2ND MIN' timer. The main area is the 'PROGRAM EDITOR' for Java, containing a code snippet for finding the second smallest element in an array. The 'INPUT' field on the right contains the numbers '5 14 2 67 92 73'. The 'OUTPUT' field at the bottom shows the result '14'.

QUESTIONS 31 / 31 completed

2ND MIN

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 class Main{
2     public static void main(String[] args){
3         Scanner sc= new Scanner(System.in);
4
5
6         int n=sc.nextInt();
7         int a[]=new int[n];
8
9         for(int i=0; i<n; i++){
10             a[i]=sc.nextInt();
11         }
12         int min=a[0], second=a[1];
13         for(int i=1; i<n; i++){
14             if(min>a[i]){
15                 second=min;
16                 min=a[i];
17             }
18             else if(a[i]<second && a[i]!=min){
19                 second=a[i];
20             }
21         }
22         System.out.println(second);
23     }
24 }
```

INPUT

5
14 2 67 92 73

OUTPUT

14

PROGRAM 30:

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and a user profile for MULE MADHU SNEHA LATHA REDDY. The left sidebar shows 'QUESTIONS 26 / 31 completed' and a 'Sort-B' timer. The main area is the 'PROGRAM EDITOR' for Java, containing a code snippet for implementing bubble sort. The 'INPUT' field on the right contains the numbers '5 5 2 4 1 3'. The 'OUTPUT' field at the bottom shows the result '1 2 3 4 5'.

QUESTIONS 26 / 31 completed

SORT-B

Write a Java program to implement bubble sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR Java Run Save

```
1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int n=sc.nextInt();
6         int[] a=new int[n];
7
8         for(int i=0; i<n; i++){
9             a[i] = sc.nextInt();
10        }
11        for(int i=0; i<n-1; i++){
12            for(int j=0; j<n-1-i; j++){
13                if(a[j]>a[j+1]){
14                    int temp=a[j];
15                    a[j] = a[j+1];
16                    a[j+1]=temp;
17                }
18            }
19        }
20        for(int i=0;i<n;i++){
21            System.out.print(a[i]+" ");
22        }
23    }
24 }
```

INPUT

5
5 2 4 1 3

OUTPUT

1 2 3 4 5

PROGRAM 31:

Sort-I

SORT-I

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         Scanner sc= new Scanner(System.in);
5
6         int n=sc.nextInt();
7         int a[]=new int[n];
8
9         for(int i=0; i<n; i++){
10             a[i]=sc.nextInt();
11         }
12         for(int i=1; i<n; i++){
13             int key=a[i];
14             int j=i-1;
15             while (j>0 && a[j] >key){
16                 a[j+1] = a[j];
17                 j--;
18             }
19             a[j+1]=key;
20         }
21         for(int i=0; i<n; i++){
22             System.out.print(a[i]+" ");
23         }
24     }
25 }
```

OUTPUT

1 2 4 5 6

5
5 2 4 6 1